

M359 University domain of discourse summary



This document can be used if you are unsure of the background to the University scenario being developed during the course. It is not referenced explicitly in the course material, but it might help clarify any confusion you have regarding the domain of discourse. It should be read in conjunction with the University scenario.

This summary is a record of some of the important details regarding this scenario. If these details had been included in the scenario description they would have obscured the core information being communicated. Some of the details given here would be part of the general knowledge that anyone could be expected to bring to this scenario. Other details are specific to this domain of discourse, and a data analyst would uncover and document this specialist information as part of the analysis process. It is not a complete description – documenting an entire domain of discourse is not possible. In addition, it is expected that a reader might disagree with the ‘reality’ being summarised here. However, during the development of the database, it is important that any disagreements relating to the meaning within the domain of discourse are identified, documented and resolved.

In effect, this summary would be part of the working documents that the analyst would develop during the refinement of the conceptual data model and their exploration of the stakeholders’ requirements. In documenting the analysis leading to the conceptual data model, the design of the database and its implementation, the information in this summary would form part of the data dictionary used throughout the development life cycle.

This summary is in the form of an Entity type catalogue and a Relationship catalogue. Sample sets of data that are to be recorded in the database, which are developed to meet the stakeholders’ requirements, are included. You will be taught how to interpret the relationship occurrence diagram notation used in this document in *Block 1* of the course material.

Entity type catalogue

Name: **Course.**

Description: This is an administrative description of the course of study that is available. Each occurrence represents a different course.

Attribute details: Each course has an identifying course code issued by the University. The course codes are of the form c1 to c9, (which limits the University to offer only nine courses each year). The title for each course is a string of characters, which may not be unique. The credit value for each course is either 30- or 60-credit points.

Sample data: **Course.**

CourseCode	Title	Credit
c2	Syntax	30
c4	Semantics	60
c7	Pragmatics	30
c5	Logic	60

Name: **Quota.**

Description: This represents the maximum number of students that can be enrolled on a course and the date that the quota number was last reviewed.
Quota is a weak entity type dependent on **Course**.

Attribute details: Each quota represents a course that has a limit to the number of students that can enrol on that course. The quota is identified by the course code of the course to which it relates. The limit is a value greater than zero. The date on which the quota was last reviewed is recorded as a standard date.

Sample data: **Quota.**

CourseCode	Limit	DateReviewed
c2	3000	10/01/07
c4	250	10/01/07
c5	250	23/04/06

Name: **Region.**

Description: This represents the administrative unit that is responsible for the management of staff and students. Each occurrence is a representation of the data for a different region.

Attribute details: The university has twelve regions, each with a unique region number (1 to 12). Each region has an address represented by a string of characters, an email address in the same format as that of the **Staff** member's email address, and a telephone number consisting of a string of digits and limited to UK national numbers.

Sample data: **Region.**

RegionNumber	Address	Telephone	EmailAddress
1	57, Longboat Street, Birmingham	0120 779165	region1@open.fake.address
2	Sumner House, Curtis Street, Manchester	0160 91659	region2@open.fake.address
3	Block 9, The Campus, Walton Hill,	01670 245365	region3@open.fake.address
4	Suite 2, Faculty Towers, Torquay	02563 13829	region4@open.fake.address
12	The Office, New York	10898 227191	region12@open.fake.address

Name: **Staff.**

Description: This represents the properties common to those persons who are under contract to a region, and who undertake certain roles for the University. Each occurrence represents a different person.

Attribute details: Each staff member is issued an identifying 4-digit staff number by the university. Staff numbers are in the range 0000 to 9999. The staff member's family name (surname) is recorded, and is not expected to be used to identify staff members. A single string of characters is used to record an address. The email address is a string containing a single '@' symbol, which separates a user name from an email address string. A staff member's telephone number consists of a string of digits. The telephone numbers are limited to UK national numbers.

Sample data:

Staff.				
StaffNumber	Name	Address	EmailAddress	Telephone
8431	French	123 Townstreet, Roadville	french@open.tutors.fake	09273 28291
7774	Redhead	17 Boot Lane, Wellesley	redhead@myownemail.com	01281 19323
5212	Heathcote	234 Rue de stad, Alphaville	heathcote2@hereIam.co.uk	08104 31221
3158	Jennings	21 Beckett Way, Mitreton	artoo@mechadroid.net	04323 342322
5324	Lai	53 Broadwalk, New York	lai2@well.parc.co.uk	03223 43652
3678	Sanderson	56 Railway Cuttings, Pottsford	helen@open.tutors.fake	08642 92753
4219	Sanderson	23 Bridge Road, Hazelton	sandy@basketball.fake.co.uk	063245 93725
6451	Clark	16 Beasley Street, Chickentown	jcc@salford.fake.gov.uk	0826 82074

Name: **Student.**

Description: This represents the properties common to all persons who are registered with the University, thus allowing them to study the courses. Each occurrence represents a different person.

Attribute details: Each student is identified by a student identifier issued by the University when the student registers. The student identifiers range from **s01** to **s99** (thus, limiting the University to 99 students). The student's family name is recorded and is not expected to be used to identify a specific student. Their home address and email address are represented in the same way as a staff member's home address and email address. The date a student registers is represented by a standard date.

Sample data: **Student.**

StudentId	Name	Address	EmailAddress	RegistrationDate
s01	Akeroyd	12 Anystreet, Anytown	Akers@tahoo.fake.com	Nov 23, 1999
s02	Thompson	8 High Street, Lowville	Pjay@thompson.fake.com	Oct 12, 2004
s05	Ellis	34 Globe Road, Smallville	G.Ellis@fake.fake.co.uk	Oct 14, 2003
s07	Gillies	29 Straight Street, Angletown	Gillies@address.fake.net	Dec 2, 1993
s09	Reeves	34 The Crescent, Curville	T.Reeves@nnet.fake.com	Dec 14, 2004
s10	Urbach	22 Hilltops, Valley Town	B.Urbach@tnet.fake.fr	May 5, 2003
s22	Bryant	84 Brook Street, Little Hacking	A.Bryant@greenmail.fake. uk	Jun 21, 2000
s38	Patel	12 Stanley Road, Pitchford	patel122@mailman.fake. uk	Oct 8, 2001
s42	Reddick	23 Kestrel Lane, Dudley	dave@belwise.fake.co.uk	Apr 23, 2002
s46	Sharp	The Farm, Lower Watley	F.Sharp@bluesky.fake.ac. uk	Feb 14, 2002
s57	Patel	4 Lower Crescent, Cinderfield	r.patel@tahoo.fake.com	Nov 5, 2000

Name: **Enrolment.**

Description: This is a record of the details of a student enrolled on a specific course. Each occurrence of an enrolment represents a student enrolled on a single course. **Enrolment** is a weak entity type, dependent on both the **Course** and **Student** entity types.

Attribute details: Each enrolment is identified by a student identifier and a course code representing the student and the course to which the enrolment relates. The date on which that student is enrolled on that course is represented by a standard date. The enrolment date must be the same as, or later than, the registration date for the student concerned.

Sample data: **Enrolment.**

StudentId	CourseCode	EnrolmentDate
s01	c4	Jan 12, 2005
s02	c5	Jan 1, 2005
s02	c7	Jun 12, 2005
s05	c2	Jun 4, 2004
s05	c7	Oct 18, 2004
s07	c4	Dec 12, 2004
s09	c4	Dec 16, 2004
s09	c2	Dec 18, 2004
s09	c7	Dec 15, 2004
s10	c7	Jun 20, 2004
s10	c4	May 5, 2004
s22	c2	Mar 15, 2002
s38	c2	Sep 18, 2003
s38	c5	Mar 9, 2004
s46	c2	Mar 1, 2002
s57	c4	Jun 30, 2001
s57	c5	Jan 20, 2003

Name: **Assignment.**

Description: This is a record of the result of a piece of assessed coursework, submitted by a student on a course. Each occurrence represents a different assignment mark for a student on a course. **Assignment** is a weak entity type dependent on **Enrolment**.

Attribute details: Each assignment is identified by the enrolment to which the assignment is related and the assignment number, which is a value between 1 and 5. (If the enrolment is for a course that has a 30-credit point value then the maximum assignment number is limited to 3.) Marks awarded to the assignment are recorded as integer values between 0 and 100. A student does not have to submit every assignment, nor do assignments necessarily get submitted in sequence.

Sample data: **Assignment.**

StudentId	CourseCode	AssignmentNumber	Mark
s01	c4	1	80
s01	c4	2	74
s02	c5	1	70
s02	c5	2	65
s02	c5	3	73
s02	c7	1	53
s02	c7	2	58
s05	c2	1	78
s05	c2	2	63
s07	c4	1	91
s09	c4	1	85
s09	c2	1	92
s09	c2	2	76
s10	c4	1	45
s22	c2	1	74
s22	c2	2	68
s38	c5	1	51
s38	c5	3	43
s57	c4	1	93
s57	c4	2	82
s57	c4	3	84
s57	c5	1	74

Name:	Examination.																												
Description:	This is a record of the result of an examination assessment taken by a student on a course. Each occurrence represents a different examination mark for a student on a course. Examination is a weak entity type dependent on Enrolment .																												
Attribute details:	Each examination result is identified by the course code, and student identifier of the course and, the student (enrolment) whose result is being recorded. The mark for the examination is an integer value in the range 0 to 100. The examination location is a string of characters representing a town in the UK where the University conducts the examinations.																												
Sample data:	Examination. <table><tr><th>StudentId</th><th>CourseCode</th><th>ExaminationLocation</th><th>Mark</th></tr><tr><td>s07</td><td>c4</td><td>Bedford</td><td>85</td></tr><tr><td>s09</td><td>c4</td><td>Taunton</td><td>63</td></tr><tr><td>s10</td><td>c4</td><td>Gateshead</td><td>27</td></tr><tr><td>s09</td><td>c7</td><td>Taunton</td><td>71</td></tr><tr><td>s05</td><td>c2</td><td>Bath</td><td>57</td></tr><tr><td>s09</td><td>c2</td><td>New York</td><td>56</td></tr></table>	StudentId	CourseCode	ExaminationLocation	Mark	s07	c4	Bedford	85	s09	c4	Taunton	63	s10	c4	Gateshead	27	s09	c7	Taunton	71	s05	c2	Bath	57	s09	c2	New York	56
StudentId	CourseCode	ExaminationLocation	Mark																										
s07	c4	Bedford	85																										
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s10	c4	Gateshead	27																										
s09	c7	Taunton	71																										
s05	c2	Bath	57																										
s09	c2	New York	56																										

Name:	SpecialCircumstancesForm.								
Description:	This is a record of the reference number of a form, submitted by a student, which explains the circumstances that might have affected their assessment performance. The date the form is received by the examination officer is also noted. Each occurrence is for a different special circumstances form.								
Attribute details:	Each special circumstances form is identified by the unique code on the form. These codes range from SC000 to SC999 . The date on which the form is submitted is held in a standard format.								
Sample data:	SpecialCircumstancesForm. <table> <tr> <th>FormNumber</th><th>DateReceived</th></tr> <tr> <td>SC458</td><td>16-04-2004</td></tr> <tr> <td>SC754</td><td>07-03-2004</td></tr> <tr> <td>SC245</td><td>19-10-2004</td></tr> </table>	FormNumber	DateReceived	SC458	16-04-2004	SC754	07-03-2004	SC245	19-10-2004
FormNumber	DateReceived								
SC458	16-04-2004								
SC754	07-03-2004								
SC245	19-10-2004								

Relationship catalogue

Name: **TelephoneTutoredBy.**

Description: A staff member may be available as a telephone tutor to any student enrolled on a specific course.

Reading: Each course is telephone-tutored by one or more staff.
Each staff (member) is a telephone tutor for zero or one course.

Additional information: An instance of **TelephoneTutoredBy** between **Staff** and **Course** instances only exists if there is an **Enrolment** instance that is related to the same **Staff** and **Course** instances by the **Tutors** and **StudiedBy** relationships.

Sample
relationship
occurrences:

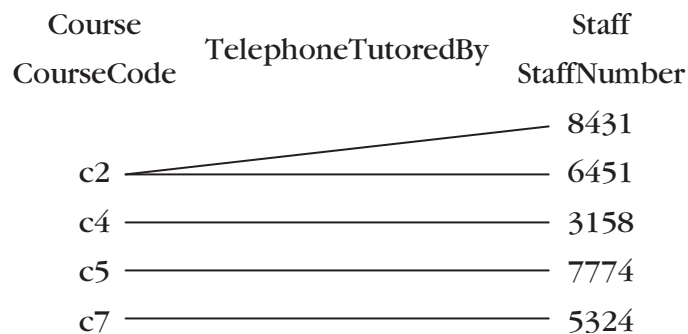


Figure 1

Name: **ContractsWith.**

Description: Regions hold employment contracts with their members of staff.

Reading: Each region contracts with zero, one or more members of staff.
Each staff (member) holds a contract with exactly one region.

Sample
relationship
occurrences:

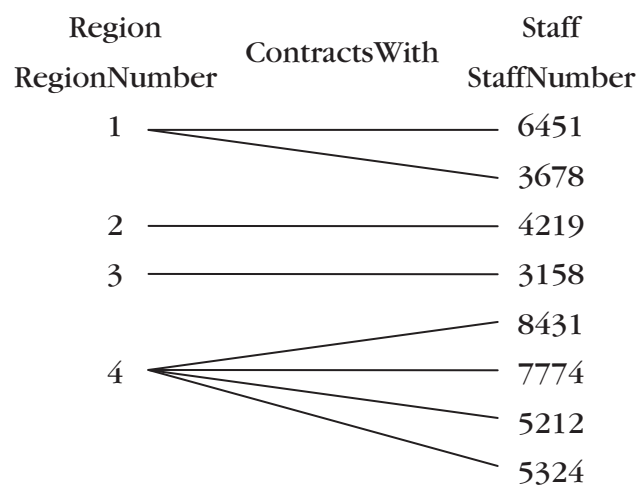


Figure 2

Name: **Manages.**

Description: The regions manage the administration for every student.

Reading: Each region manages zero, one or more students.
Each student is managed by exactly one region.

Sample
relationship
occurrences:

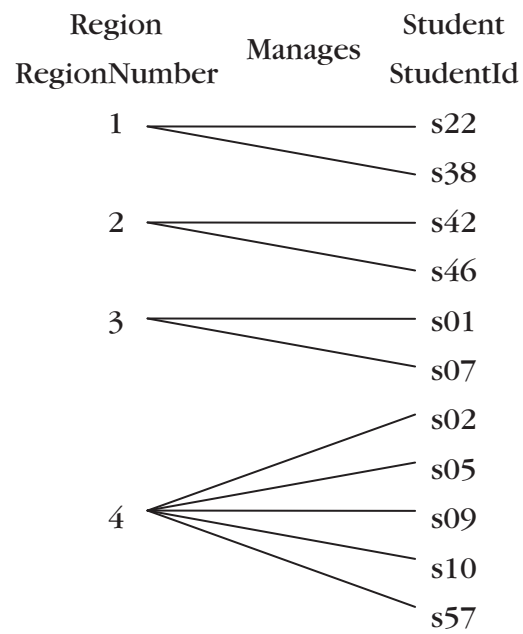


Figure 3

Name: **EnrolledIn.**

Description: Each student takes specific courses.

Reading: Each student is enrolled in zero, one or more enrolments.
Each enrolment enrolls exactly one student.

Additional information: The number of instances of the **EnrolledIn** relationship, which an instance of **Student** can be involved in, is limited by the restriction that the credit total of the courses, to which the **Enrolment** instances relate, must not exceed 180 points.

Sample relationship occurrences:

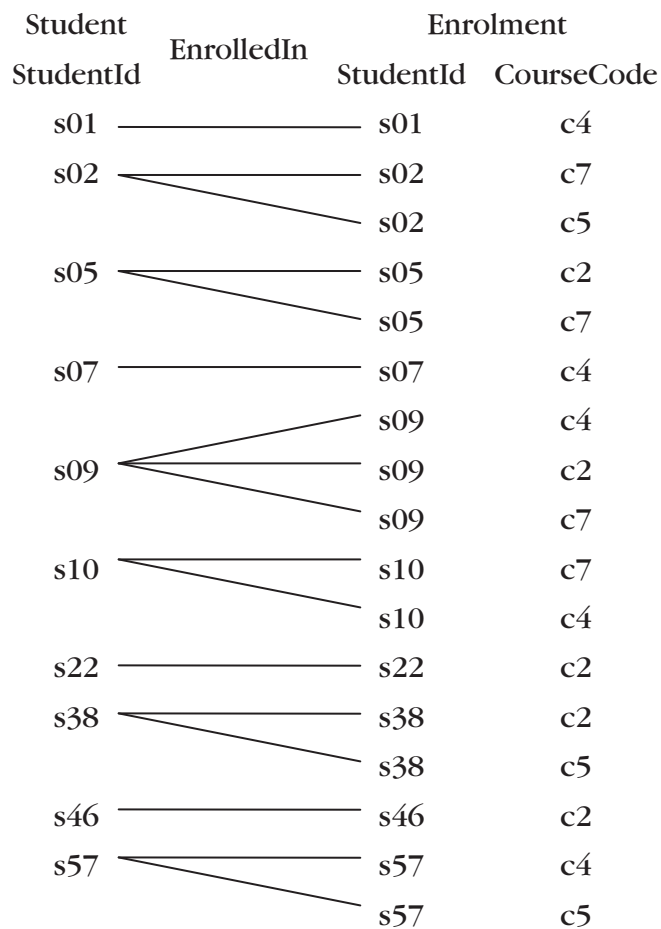


Figure 4

Name: **StudiedBy.**

Description: Each student enrolls on specific courses.

Reading: Each course is studied by zero, one or more enrolments.
Each enrolment is to study exactly one course.

Sample
relationship
occurrences:

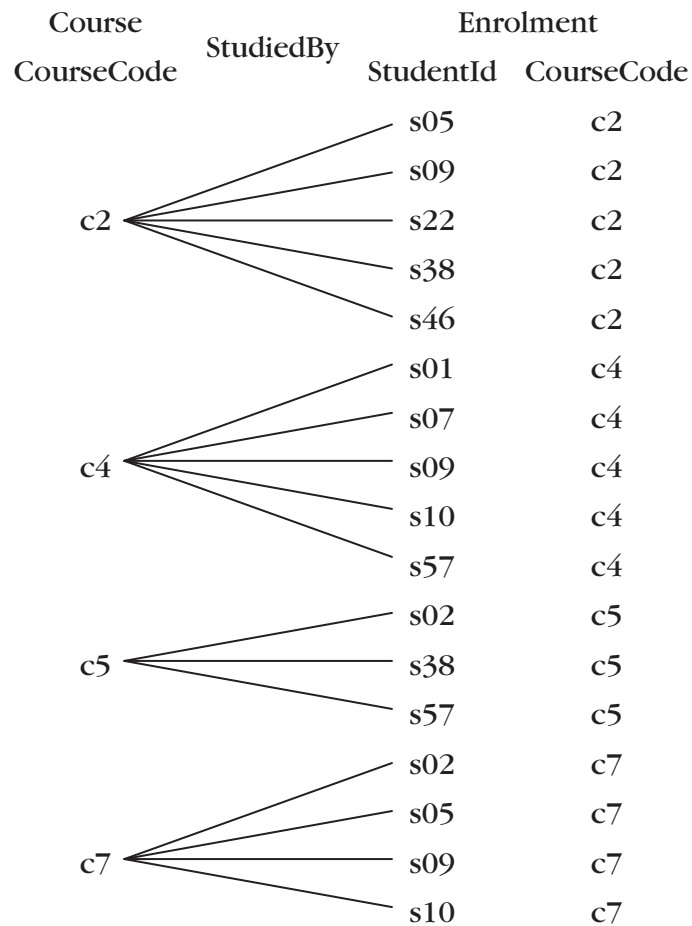


Figure 5

Name: **Tutors.**

Description: A staff member acts as a tutor for one or more students taking specific courses.

Reading: Each staff (member) tutors zero, one or more enrolments.
Each enrolment is tutored by zero or one staff (member).

Additional information: An instance of the **Tutors** relationship between instances of **Staff** and **Enrolment** can only exist if there is a **Student** instance related to the **Enrolment** instance and to the same **Region** as the **Staff** instance.

Sample
relationship
occurrences:

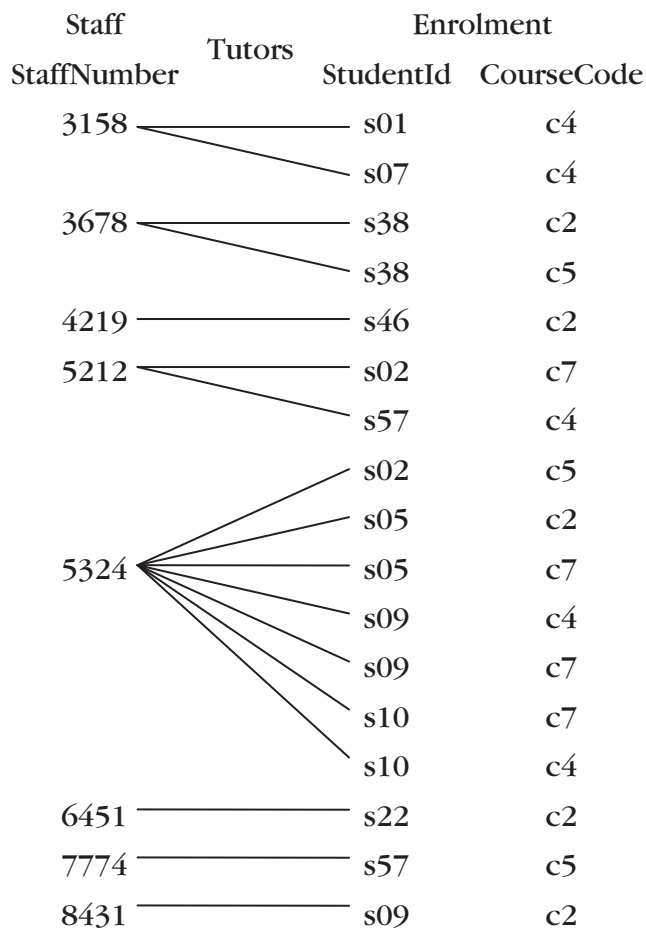


Figure 6

Name: **Submits.**

Description: A student enrolled on a course has their assignment results recorded.

Reading: Each enrolment submits zero, one or more assignments.

Each assignment is submitted by exactly one enrolment.

Additional information: The number of instances of the **Submits** relationship for each **Enrolment** instance is limited by the credit value of the course to which the enrolment is related.

Sample
relationship
occurrences:

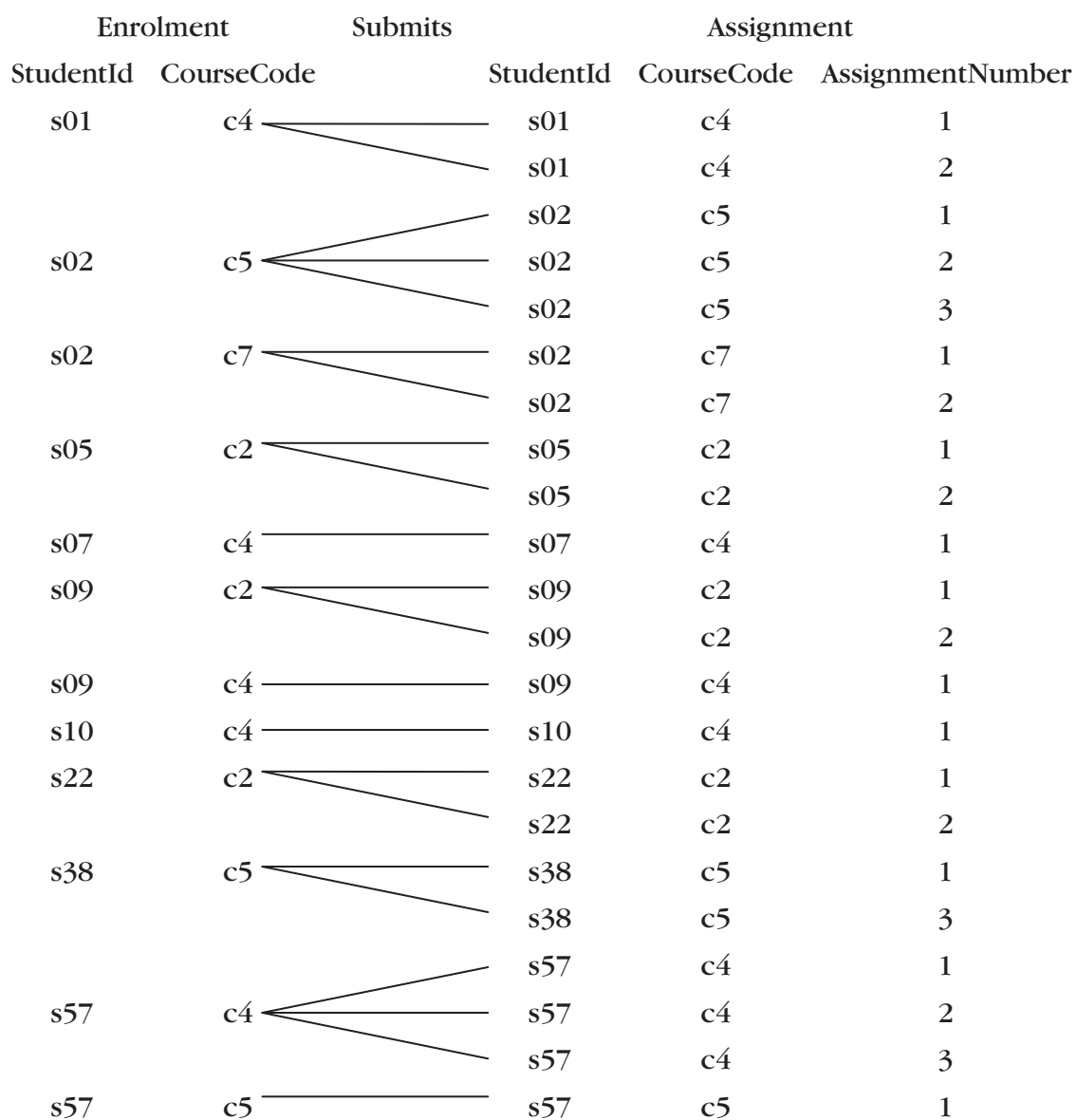


Figure 7

Name: **Takes.**

Description: A student enrolled on a course has an examination recorded.

Reading: Each enrolment takes zero or one examinations.
Each examination is taken by exactly one enrolment.

Sample
relationship
occurrences:

Enrolment		Takes	Examination	
StudentId	CourseCode		StudentId	CourseCode
s07	c4	—————	s07	c4
s09	c4	—————	s09	c4
s10	c4	—————	s10	c4
s05	c2	—————	s05	c2
s09	c2	—————	s09	c2

Figure 8

Name: **LimitedBy.**

Description: A course can have a limit on the number of students.

Reading: Each course is limited by zero or one quota.
Each quota is the limit for exactly one course.

Sample
relationship
occurrences:

Course	LimitedBy	Quota
CourseCode		CourseCode
c2	—————	c2
c4	—————	c4
c5	—————	c5

Figure 9

Name: **ExamIncludes.**

Description: A student may include a special circumstances form with each examination.

Reading: Each examination includes zero or one special circumstances form.
Each special circumstances form is included with zero or one examination.

Additional information: **ExamIncludes** and **AssignmentIncludes** are exclusive relationships with respect to the special circumstances form.

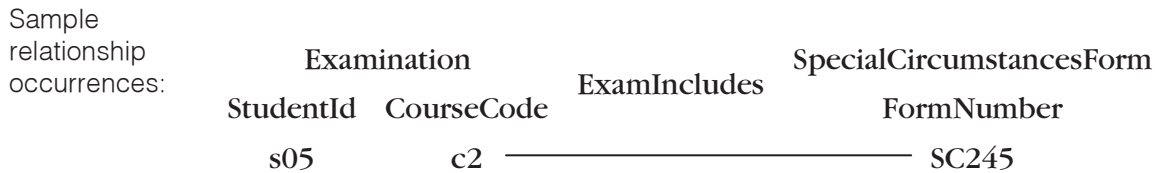


Figure 10

Name: **AssignmentIncludes.**

Description: A student may include a special circumstances form with each assignment.

Reading: Each assignment includes zero or one special circumstances form.
Each special circumstances form is included with zero or one assignment.

Additional information: **ExamIncludes** and **AssignmentIncludes** are exclusive relationships with respect to the special circumstances form.

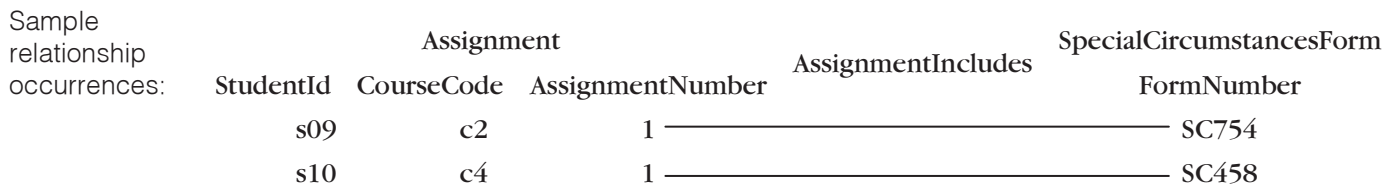


Figure 11